Project Title	**📊 App User Behavior Segmentation Using Unsupervised Machine Learning**
Skills take away From This Project	• **Data Cleaning & Preprocessing** • **Exploratory Data Analysis (EDA) for behavioral insights** • **Feature Engineering & Selection for clustering models** • **Data Scaling & Normalization (StandardScaler)** • **Unsupervised Machine Learning** • **Dimensionality Reduction using PCA** • **Cluster Profiling & Interpretation** • **Model Evaluation without Labels** • **Data Visualization (Matplotlib, Seaborn)**
Domain	• **User Behavior Analytics,** • **Machine Learning**

Problem Statement:

Applications generate large amounts of user activity data, but understanding user behavior without predefined labels is difficult.

This project focuses on analyzing app usage data using unsupervised machine learning techniques.

The objective is to group users based on similar behavior and engagement patterns.

By identifying different user segments, the project highlights high-engagement and at-risk users.

These insights help improve user engagement, retention strategies, and overall product performance.

💼 Business Use Cases

1. **User Segmentation for Targeted Marketing**
   Identify different user groups based on engagement levels to run personalized marketing campaigns and improve conversion rates.

2. **Churn Risk Identification and Retention Planning**
   Detect low-engagement and inactive users early, enabling proactive retention strategies such as offers, reminders, or improved onboarding.

3. **Personalized User Experience**
   Customize app features, notifications, and recommendations for different user segments to enhance overall user satisfaction.

4. **Product Feature Optimization**
   Analyze how different user groups interact with app features to improve or prioritize features based on actual user behavior.

5. **Data-Driven Business Decision Making**
   Support strategic decisions related to product development, customer support, and monetization using insights derived from user behavior patterns.

🔍 Approach

1. Data Collection

Collected a large-scale app user behavior dataset containing user demographics, device details, engagement metrics, session activity, and interaction patterns.

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2. Data Understanding

Explored the dataset to understand feature definitions, data types, value ranges, and overall data quality using basic statistics and summaries.

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3. Data Cleaning

Handled missing values using appropriate imputation techniques and ensured consistency in numerical and categorical features to make the data suitable for analysis.

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4. Feature Selection

Identified and selected relevant numerical features such as session frequency, session duration, engagement score, and activity metrics that influence user behavior.

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5. Data Scaling

Applied feature scaling using StandardScaler to normalize the data, ensuring that all features contribute equally to the clustering process.

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6. Clustering Model Selection

Chose K-Means clustering as the unsupervised learning algorithm due to its effectiveness in grouping users based on behavioral similarity.

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7. Optimal Cluster Identification

Used the Elbow Method to determine the optimal number of clusters by analyzing inertia values and selecting the point where improvement stabilizes.

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8. Cluster Assignment

Trained the K-Means model with the optimal number of clusters and assigned a cluster label to each user in the dataset.

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9. Cluster-Level User Identification & Profiling

Identified which customers belong to each cluster using user IDs and analyzed their behavioral characteristics.

Tasks performed in this step:

- Extracted user lists for each cluster (High, Moderate, Low, Occasional users)
- Counted the number of customers in each group
- Analyzed average engagement, session behavior, and churn risk per cluster
- Created customer-level profiles to understand behavior patterns within each group

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10. Business Insight Generation & Customer Action Mapping

Mapped each user cluster to specific business actions by understanding who the customers are in each group.

Tasks performed in this step:

- Identified high-value customers for loyalty and premium offers
- Flagged low-engagement and at-risk customers for retention campaigns
- Designed personalized engagement strategies for moderate users
- Supported marketing, product, and retention decisions based on customer segments
- Enabled customer-level targeting instead of generic campaigns

📊 Project Results

1. Successful User Segmentation

The project successfully applied **K-Means clustering** on app user behavior data and segmented 50,000 users into **four distinct behavioral groups** without using any labeled target variable.

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2. Identified User Clusters

The clustering results produced the following user segments:

- **Cluster 0 – High Engagement Users**
  Users in this cluster showed high session frequency, longer session duration, high engagement scores, and low churn risk.

- **Cluster 1 – Moderate Engagement Users**
  These users demonstrated consistent but average usage patterns with moderate

engagement levels and balanced activity.

- **Cluster 2 – Low Engagement / At-Risk Users**
  This group consisted of users with low activity, infrequent logins, shorter sessions, and higher churn risk indicators.

- **Cluster 3 – Occasional Users**
  Users in this cluster interacted with the app irregularly, showing sporadic usage and lower feature interaction.

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3. Clear Behavioral Separation

Visualization using **PCA** confirmed clear separation between user clusters, validating that the selected features effectively captured behavioral differences.

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4. Customer-Level Identification

Each user in the dataset was successfully assigned to a cluster, enabling the identification of **exact customers belonging to each engagement group** for targeted actions.

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5. Business-Actionable Insights

The clustering results provided actionable insights such as:

- Identifying high-value users for loyalty programs
- Detecting churn-risk users for retention strategies
- Enabling personalized engagement plans for different user segments

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6. Scalable and Real-World Ready Outcome

The project demonstrated the ability to handle large-scale real-world-like data and deliver meaningful insights suitable for business decision-making and further dashboard integration.

📊 Project Evaluation Metrics

- **Data Quality & Preprocessing Accuracy – 20%**
  Measures how effectively missing values, inconsistencies, and scaling were handled to ensure reliable clustering.

- **Feature Selection & Scaling – 15%**
  Evaluates the relevance of selected features and proper normalization for unsupervised learning.

- **Cluster Quality & Separation – 25%**
  Assesses how well users are grouped into distinct clusters based on behavioral similarity.

- **Cluster Interpretability – 15%**
  Checks how clearly each user segment can be explained in terms of engagement and activity.

- **Business Relevance & Actionability – 15%**
  Evaluates the usefulness of clusters for real-world actions like retention and personalization.

- **Scalability & Performance – 10%**
  Measures the model's ability to handle large datasets efficiently.

Data Set:
https://drive.google.com/file/d/1YVwuLd4zgrqM2i10XGf3OG2ean86RDuX/view?usp=sharing

Data Set Explanations :

Dataset_Explanations.

References:

Project Live Evaluation Metrics	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) Guide
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Project Orientation (Tamil)	Project Orientation Session : App …
Project Orientation (English)	

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:30PM to 4:30PM)

Booking link : **Project Doubt Session [DS/AIML]**

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:
For DS and AIML
Monday-Saturday (05:30PM to 07:00PM)

- ***Booking link :*** https://forms.gle/1m2Gsro41fLtZurRA

Evaluation Metrics : Project Live Evaluation Metrics

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